



Alert Triage with Automation

1. Introduction

Alert triage is a critical SOC function that involves validating, prioritizing, and investigating security alerts to determine whether they represent genuine threats or false positives. Modern SOC's rely on automation and threat intelligence integration to improve triage efficiency and reduce response time.

This report demonstrates alert triage using:

- **Wazuh Inc. Wazuh** (Alert Detection)
- **VirusTotal VirusTotal** (Threat Intelligence Validation)
- **StrangeBee TheHive** (Incident Case Management)

The objective was to simulate a suspicious file download alert and automate validation using external threat intelligence.

2. Objective

The objective of this task was to:

- Analyze and validate a suspicious alert.
 - Prioritize based on risk level.
 - Automate hash reputation checks.
 - Create structured documentation in case management system.
 - Reduce manual validation time.
-

3. Alert Scenario

A security alert was generated in Wazuh indicating a suspicious file download on an internal workstation.

Alert Details:

Alert ID	Description	Source IP	Priority	Status
005	Suspicious File Download	192.168.1.102	High	Open



4. Alert Triage Process

4.1 Step 1 – Initial Alert Review

- Reviewed alert metadata.
- Verified timestamp and source system.
- Identified downloaded file hash.
- Assessed risk based on alert severity.

The alert was classified as **High Priority** due to potential malware infection risk.

4.2 Step 2 – Threat Intelligence Validation

The file hash extracted from Wazuh was submitted to VirusTotal for reputation analysis.

VirusTotal Findings:

- Multiple antivirus engines flagged file as malicious.
- Hash associated with known trojan activity.
- IP linked to suspicious hosting provider.

This confirmed that the alert was a **True Positive**.

4.3 Step 3 – Automated Validation Using TheHive

TheHive was configured to automatically:

- Extract file hash.
- Query VirusTotal API.
- Attach reputation results to case.
- Assign severity based on result.

Automation reduced validation time significantly.

5. Case Creation in TheHive

Once validated, an incident case was created.

Case Information:

- **Title:** Suspicious File Download – High Risk
-



- **Severity:** High
- **Source IP:** 192.168.1.102
- **Indicators:** Malicious file hash
- **Status:** Under Investigation

The automation engine populated investigation notes with VirusTotal findings.

6. Automated Validation Summary

A suspicious file download alert was generated in Wazuh and classified as high priority. The file hash was automatically submitted to VirusTotal through TheHive integration. The hash was flagged as malicious by multiple engines. The alert was confirmed as a true positive and escalated for containment and further investigation.

7. Containment Actions

Following validation:

- Endpoint temporarily isolated.
- Malicious file quarantined.
- Network traffic monitored for lateral movement.
- Additional IOC sweep conducted across endpoints.

Containment actions were documented within TheHive case.

8. Analysis

This exercise demonstrated:

- Importance of automated hash validation.
- Reduced manual workload.
- Faster confirmation of malicious artifacts.
- Improved accuracy of alert triage.
- Integration benefits between SIEM and case management.

Automation significantly reduced Mean Time to Respond (MTTR).



9. Challenges Identified

- False positives still require contextual analysis.
 - Threat intelligence API rate limits.
 - Need for periodic playbook optimization.
 - Importance of human verification for critical incidents.
-

10. Recommendations

- Integrate additional threat intelligence feeds.
 - Implement automated endpoint isolation.
 - Use sandbox analysis for unknown files.
 - Regularly tune alert severity thresholds.
 - Monitor false positive rate metrics.
-

11. Conclusion

This task successfully demonstrated structured alert triage combined with automation and threat intelligence integration. By leveraging Wazuh for detection, VirusTotal for validation, and TheHive for case management, the SOC workflow became more efficient and reliable. Automated validation significantly reduced investigation time while maintaining accuracy, strengthening overall SOC performance.